

DYNAMICAL SYSTEMS - MA3081

Solutions to Exercise Sheet 4

Exercise 4.1 (Conjugacy for similar matrices).

Consider the two dynamical systems generated by

$$x' = Ax \quad \text{and} \quad y' = By$$

for some matrices $A, B \in \mathbb{R}^{n \times n}$.

Show that the following are equivalent:

- (i) The flows are linearly conjugate, i.e. the conjugating homeomorphism is a linear map.
- (ii) The flows are C^1 -conjugate.
- (iii) The matrices A and B are similar, i.e. $B = S^{-1}AS$.

Solution.

We show (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Rightarrow$ (i).

(i) $\Rightarrow$ (ii) is clear, since every linear map is C^1 .

(ii) $\Rightarrow$ (iii): Assume that the flows, let us call them ϕ and ψ , are C^1 -conjugate, i.e. there is some $h \in C^1$ with $h(\phi_t(x)) = \psi_t(h(x))$ for every $t \in \mathbb{R}, x \in \mathbb{R}^n$. We differentiate $h(\phi_t(x)) = \psi_t(h(x))$ with respect to x and obtain

$$Dh(\phi_t(x))D\phi_t(x) = D\psi_t(h(x))Dh(x).$$

Taking $x = 0$ and the explicit formulas $\phi_t(x) = e^{At}x, \psi_t(x) = e^{Bt}x$ then yields

$$Dh(0)e^{At} = e^{Bt}Dh(0).$$

Now differentiate with respect to t and evaluate at $t = 0$ to get

$$Dh(0)A = B Dh(0).$$

Since h is a diffeomorphism, $Dh(0)$ is invertible and consequently

$$A = Dh(0)^{-1}B Dh(0),$$

i.e. A and B are similar.

(iii) $\Rightarrow$ (i): Let A and B be similar with $B = S^{-1}AS$. We show that $h(x) = Sx$ conjugates the flows. Observe that from the similarity we also obtain $B^k = S^{-1}A^kS$ and consequently $e^{Bt} = S^{-1}e^{At}S$. This already shows that $h(e^{Bt}x) = Se^{Bt}x = e^{At}Sx = e^{At}h(x)$, i.e. e^{At} and e^{Bt} are linearly conjugate. $\square$

Exercise 4.2 (Topological conjugacy of linear systems I).

We consider the two differential equations

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} -1 & -1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

- Rewrite the differential equations in polar coordinates $x = r \cos(\theta)$, $y = r \sin(\theta)$.
- Sketch the phase portraits around the origin.
- Show that the two dynamical systems are conjugate. Are they also C^1 -conjugate?

Hint: Find a homeomorphism that keeps the circle $\{r = 1\}$ invariant.

Solution.

- We rewrite the systems in polar coordinates (r, θ) , defined by $x = r \cos(\theta)$, $y = r \sin(\theta)$. Differentiating $r^2 = x^2 + y^2$ we obtain $rr' = xx' + yy'$ so $r' = \frac{1}{r}(xx' + yy')$ and from $\theta = \arctan(\frac{y}{x})$ we obtain $\theta' = \frac{xy' - yx'}{r^2}$.

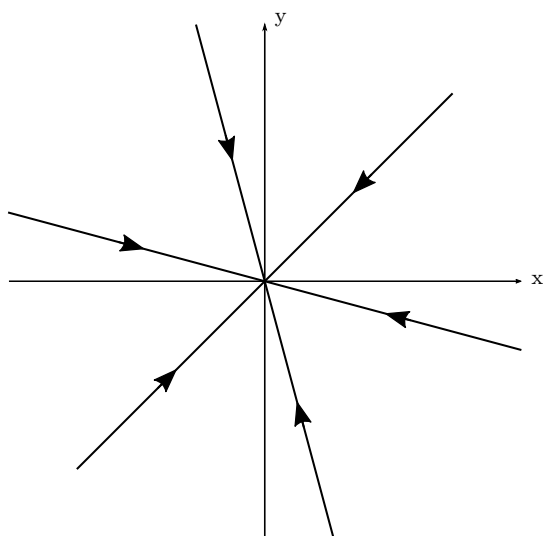
Hence the first system is transformed into

$$\begin{aligned} r' &= -r, \\ \theta' &= 0 \end{aligned}$$

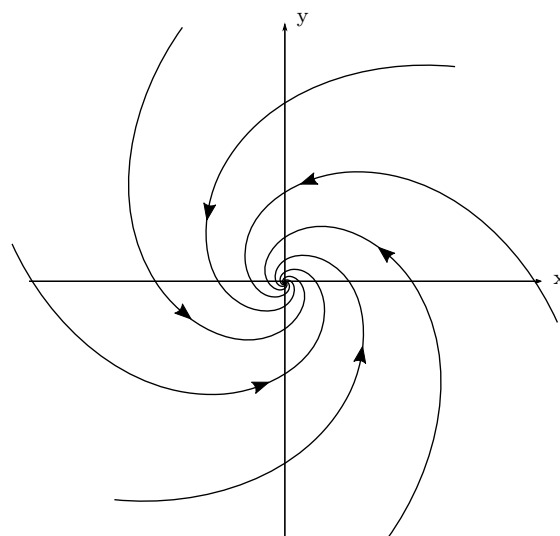
and the second system into

$$\begin{aligned} r' &= -r, \\ \theta' &= 1. \end{aligned}$$

- The two differential equations in polar coordinates easily allow to draw the phase portraits. Since $\theta' = 0$ in the first system, orbits are straight lines to the origin. Since $\theta' = 1$ in the second system, here orbits spiral counterclockwise.



$$\begin{aligned} r' &= -r \\ \theta' &= 0 \end{aligned}$$



$$\begin{aligned} r' &= -r \\ \theta' &= 1 \end{aligned}$$

- c) For the two systems of differential equations in polar coordinates from a) we immediately find explicit solutions to some initial conditions (r_0, θ_0) at time $t = 0$.

For the first system we get

$$\phi_t(r_0, \theta_0) = \begin{pmatrix} r(t) \\ \theta(t) \end{pmatrix} = \begin{pmatrix} r_0 e^{-t} \\ \theta_0 \end{pmatrix}$$

and for the second system

$$\psi_t(r_0, \theta_0) = \begin{pmatrix} r(t) \\ \theta(t) \end{pmatrix} = \begin{pmatrix} r_0 e^{-t} \\ \theta_0 + t \end{pmatrix}.$$

Let us now construct the homeomorphism h by identifying two trajectories that go through the same point on the circle $\{r = 1\}$. An initial condition at $(r, \theta) = (1, \theta_0)$ evolves to $(r(t), \theta(t)) = (e^{-t}, \theta_0)$ and $(r(t), \theta(t)) = (e^{-t}, \theta_0 + t)$ respectively. Hence h can keep the radial component the same. Note that we have $\ln(r(t)) = -t$ for the considered trajectories. Thus, we can use the following function h to map orbits of the first system to those of the second:

$$h(r, \theta) = (r, \theta - \ln(r)).$$

Note that this is only well-defined for $r > 0$, but since we want the equilibrium at the origin to get mapped to itself, we extend h by $h(0, \theta) = (0, \theta)$. The function h is clearly continuous at any point beside the origin. Since h maintains the euclidean norm (radial component stays the same) it is also continuous at the origin. The same is true for its inverse $h^{-1}(r, \theta) = (r, \theta + \ln(r))$.

Using the explicit solutions above we can also check that h indeed gives the right conjugacy:

$$\begin{aligned} h(\phi_t(r, \theta)) &= h(re^{-t}, \theta) = (re^{-t}, \theta - \ln(re^{-t})) \\ &= (re^{-t}, \theta + t - \ln(r)) = \psi_t(r, \theta - \ln(r)) = \psi_t(h(r, \theta)). \end{aligned}$$

We have seen in Exercise 4.1 that two linear ODE systems are C^1 -conjugate if and only if the corresponding matrices are similar. This is not the case so the two systems are not C^1 -conjugate.

□

Exercise 4.3 (Topological conjugacy of linear systems II).

We consider the two differential equations

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

Show that the two dynamical systems are topologically conjugate using a homeomorphism of the form $h(x, y) = (x + f(y), y)$ for some $f: \mathbb{R} \rightarrow \mathbb{R}$.

Hint: Derive and solve a (differential) equation for the unknown function f .

Solution.

We want to find a homeomorphism h of the form $h(x, y) = (x + f(y), y)$ for some function $f : \mathbb{R} \rightarrow \mathbb{R}$, that satisfies

$$h(\phi_t(x, y)) = \psi_t(h(x, y)). \quad (4.1)$$

Here $\phi_t(x, y)$ and $\psi_t(x, y)$ denote the flows of the linear differential equations $\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$ and $\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$ respectively and they are given by

$$\phi_t(x, y) = (e^{2t}x, e^{2t}y) \quad \text{and} \quad \psi_t(x, y) = (e^{2t}x + te^{2t}y, e^{2t}y).$$

There are different ways to find f , so that the above equation (4.1) holds; let us have a closer look on two approaches:

- (i) Let us assume that h is (piecewise) diffeomorphic and conjugates the flows ϕ_t and ψ_t . (Even if h may not be differentiable at every point in the end, this approach gives great insight how f can look like).

By Proposition 4.3 it holds that

$$\begin{aligned} Dh(x, y) \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} &= \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix} h(x, y) \\ \Leftrightarrow \begin{pmatrix} 1 & f'(y) \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} &= \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} x + f(y) \\ y \end{pmatrix}. \end{aligned}$$

The equation of the second component is trivial, but in the first component, we obtain a differential equation, namely

$$\begin{aligned} 2x + 2f'(y)y &= 2x + 2f(y) + y \\ \Leftrightarrow f'(y)y &= f(y) + \frac{y}{2}. \end{aligned} \quad (4.2)$$

As long as $y \neq 0$ this is equivalent to

$$f'(y) = \frac{f(y)}{y} + \frac{1}{2}$$

which can be solved using variation of constants.

To solve the differential equation for f one may also take (4.2) and differentiate (again assuming enough regularity of f) to get

$$\begin{aligned} f''(y)y + f'(y) &= f'(y) + \frac{1}{2} \\ \Leftrightarrow f''(y)y &= \frac{1}{2}. \end{aligned}$$

Integrating twice (and neglecting integration constants, since we are only interested in finding one solution) we obtain $f(y) = \frac{1}{2}y \ln |y|$. Since $\lim_{y \rightarrow 0} \frac{1}{2}y \ln |y| = 0$, we can extend f to a continuous function on $\mathbb{R}$ by setting $f(0) = 0$.

This choice of f yields the function $h(x, y) = (x + \frac{1}{2}y \ln |y|, y)$, which is bijective with continuous inverse $h^{-1}(x, y) = (x - \frac{1}{2}y \ln |y|, y)$. Even though h is not differentiable (at 0) one quickly verifies that h indeed fulfills equation (4.1), i.e. we have shown topological conjugacy (but not C^1 -conjugacy).

- (ii) For the second approach we do not need to assume differentiability of h . Taking the explicit representations of ϕ_t and ψ_t and plugging them into (4.1), yields

$$(e^{2t}x + f(e^{2t}y), e^{2t}y) = (e^{2t}(x + f(y)) + te^{2t}y, e^{2t}y)$$

In the second component we have the same entry so it remains to find a function f such that

$$\begin{aligned} e^{2t}x + f(e^{2t}y) &= e^{2t}(x + f(y)) + te^{2t}y \\ \Leftrightarrow f(e^{2t}y) &= e^{2t}f(y) + te^{2t}y. \end{aligned}$$

Let us substitute $e^{2t} = s$ which gives

$$f(sy) = sf(y) + \frac{1}{2} \ln(s)sy. \quad (4.3)$$

Now, set $y = 1$ to get

$$f(s) = sf(1) + \frac{1}{2} \ln(s)s.$$

Since $s = e^{2t} > 0$ and we also want f for negative arguments, we next take $y = -1$ in (4.3) to get (with $u = -s < 0$)

$$f(-s) = sf(-1) - \frac{1}{2} \ln(s)s \Leftrightarrow f(u) = -uf(-1) + \frac{1}{2} \ln(-u)u.$$

One checks that the required functional equations are satisfied for any choice of $f(1)$ and $f(-1)$ so for simplicity we can assume them to be 0. In total this also gives the function $f(y) = \frac{1}{2}y \ln |y|$ (for $y \neq 0$, extend as above).

□

Exercise 4.4 (Topological conjugacy of nonlinear systems).

Show that the two dynamical systems induced by

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} -\sin(x) + 3y^3 \\ x^2 - y \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} -x + y(y-1) \\ e^x - e^y \end{pmatrix}$$

are locally conjugate around the origin.

Solution.

Both systems have an equilibrium point at the origin.

We linearize at $(0, 0)$ and obtain $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$ for the Jacobian of the first system and $\begin{pmatrix} -1 & -1 \\ 1 & -1 \end{pmatrix}$ for the second one. Hence both equilibria are hyperbolic.

By the Hartman-Grobman Theorem the systems are locally topologically conjugate to their respective linearizations. In Exercise 4.2, however, we have shown that these two linear systems are topologically conjugate. Since topological conjugacy is an equivalence relation, we can conclude that the two nonlinear systems have to be locally topologically conjugate as well. □